Supplementary

Related work

The Segment Anything Model (SAM) [1] is the first foundation model in image segmentation. However, the applicability of SAM to medical image segmentation remains limited due to the significant differences between natural images and medical images. SAM is a promptable segmentation method that requires points or bounding boxes to specify the segmentation targets. This resembles conventional interactive segmentation methods [2]–[5] but SAM has better generalization ability, while existing deep learning-based interactive segmentation methods focus mainly on limited tasks and image modalities. Several studies have shown that SAM can fail in typical medical image segmentation tasks [6]–[13] and other challenging scenarios [14]–[17], especially when targets have weak boundaries. This motivates us to develop a universal segmentation tool that can segment a wide range of anatomical structures and medical imaging modalities. While concurrent studies [18]–[20] also tried to adapt SAM for the medical domain, our work stands out as the first pre-trained model specifically for general medical image segmentation tasks.

TABLE 1
Computed Tomography (CT) image datasets. Datasets marked with $*$ denote external validation sets and the remaining datasets are used for internal validation.

Dataset Name	Modality	Segmentation Targets	# of scans
AbdomenCT-1K [21], [22]	CT	Liver, kidneys, pancreas, spleen	1056
AMOS-CT [23]	CT	Abdominal organ	200
AutoPET [24]	PET-CT	Whole-body tumor	900
COVID-19 Seg. Challenge [25], [26]	CT	COVID-19 infections	199
COVID-19-CT-Seg [27]	CT	COVID-19 infections, left lung, and right lung	20
GLIS-RT [28]	CT	Head tumor	75
HCC-TACE-Seg* [29], [30]	CT	Liver cancer	70
HECKTOR [31]	PET-CT	Head and neck tumor	524
INSTANCE [32]	CT	Hematoma	100
KiPA [33], [34]	CT	Kidney, tumor, renal artery, renal vein	70
KiTS [35]	CT	Kidney, tumor, cyst	489
Lymph Nodes [36], [37]	CT	Lymph nodes	176
MMWHS-CT [38]–[41]	CT	Heart anatomies	20
MSD-Colon Tumor [42]	CT	Colon tumor	126
MSD-Hepatic Tumor [42]	CT	Hepatic tumor	303
MSD-Lung Tumor [42]	CT	Lung tumor	96
MSD-Pancreas [42]	CT	Pancreas, pancreas tumor	281
MSD-Spleen [42]	CT	Spleen	61
NSCLC Pleural Effusion [29], [43], [44]	CT	Pleural effusion	78
NSCLC Radiogenomics [45]	CT	Lung Tumor	88
ORG [46]	CT	Whole-body organs	140
SegTHOR [47]	CT	Esophagus, heart, aorta, trachea	40
StructSeg* [48]	CT	Nasopharyngeal cancer and lung cancer, with OAR and GTV	50
TotalSegmentator [49]	CT	Whole body organs	1204
WORD* [50]	CT	Abdominal organs	150

TABLE 2
Magnetic Resonance (MR) image datasets. Datasets marked with $*$ denote external validation sets and the remaining datasets are used for internal validation.

Dataset Name	Modality	Segmentation Targets	# of scans
ACDC* [51]	MR	Heart anatomies	150
AMOS-MR [23]	MR	Abdominal organ	40
ATLAS R2.0 [52]	MR-T1	Brain stroke	1271
Brain Tumor Dataset Figshare [53], [54]	MR-T1ce	Brain tumor	233
Brain TR-GammakKnife [55]	MR	Brain lesion	47
BraTS [56]–[60]	MR-T1, MR-T1CE, MR-T2, MR-FLAIR	Brain tumor	1251
CC-Tumor Heterogeneity* [61]	MR	Cervical cancer	7
CHAOS* [62]	MR-T1, MR-T2	Liver, kidney, spleen	60
crossMoDA [63]	MR	Brain tumor	227
FeTA [63]	MR-Fetal	Brain tissues	160
HaN-Seg* [64]	MR	Head organs	42
ISLES [65]	MR-DWI, MR-ADC, MR-FLAIR	Ischemic stroke lesion	180
I2CVB [66]	MR-T2, MR-DWI	Prostate	19
Meningioma-SEG-CLASS [67]	MR-T1ce, T2-FLAIR	Tumor (meningioma)	191
MMs [68]	MR	Heart anatomies	150
MMWHS-MR [38]–[41]	MR	Heart anatomies	20
MSD-Heart [69]	MR	Left atrial	30
MSD-Prostate [42]	MR-ADC, MR-T2	Prostate	48
NCI-ISBI [70]	MR-ADC, MR-T2	Prostate	48
PI-CAI [71]	MR-bp	Prostate cancer	1584
PPMI [72]	MR-T1	Brain regions of Parkinson patients	1130
PROMISE [73]	MR-T2	Prostate	50
Qin-Prostate-Repeatability [42], [74]	MR	Left atrium	30
QUBIQ* [75]	MR	Prostate	52
Spine [76]	MR	Vertebrae	172
WMH [77]	MR-T1, MR-FLAIR	White matter hyper-intensities	60

TABLE 3
Chest X-ray (CXR), Mammography, Optical Coherence Tomography (OCT), and Ultrasound image datasets. Datasets marked with $*$ denote external validation sets and the remaining datasets are used for internal validation.

Dataset Name	Modality	Segmentation Targets	# of images
Chest Xray Masks and Labels [78] [79]	CXR	Lung	704
Chest X-Ray (Pneumothorax) [80] [81]	CXR	Pneumothorax	2668
COVID-19 Radiography* [82] [83]	CXR	Lung, COVID-19 infection, lung opacity, viral pneumonia	21165
COVID-QU-Ex [84] [82] [85] [83] [86]	CXR	Lung, COVID infection	5826
JSRT [87]	CXR	Lung, heart	307
Lung [88] [89] [90]	CXR	Lung	30330
QaTa-COV19 [91]	CXR	COVID-19 infection	9258
CDD-CESM [29] [92] [93]	Mammography	Abnormal findings in breast tissue	1233
Intraretinal Cystoid Fluid [94] [95]	OCT	Cystoid macular edema	1460
OCT Images (DME) [96]	OCT	Diabetic macular edema	610
AbdomenUS [97]	Ultrasound	Gallbladder, kidney, liver, spleen, vessel	60
Breast Cancer [98]	Ultrasound	Benign & malignant breast lesion	647
CAMUS [99]	Ultrasound	Left ventricle endocardium, left atrium	21232
CT2USforKidneySeg [100]	Ultrasound	Kidney	4586
FH-PS-AOP [101]	Ultrasound	Pubic symphysis, fetal head	4000
HC [102]	Ultrasound	Fetal head circumference	999
TN-SCUI [103]	Ultrasound	Benign & malignant thyroid nodules	3644
Nerve [104]	Ultrasound	Brachial plexus	5635

TABLE 4
Dermoscopy, Endoscopy, Fundus, and Pathology image datasets. Datasets marked with $*$ denote external validation sets and the remaining datasets are used for internal validation.

Dataset Name	Modality	Segmentation Targets	# of images
ISIC [105] [106] [107]	Dermoscopy	Skin cancer	3694
UWaterloo Skin Cancer* [108]	Dermoscopy	Skin cancer	206
BKAI-IGH NeoPolyp [109]	Endoscopy	Polyp	1000
CholecSeg8k [110] [111]	Endoscopy	Surgical tools and abdominal tissues	8080
Kvasir* [112]	Endoscopy	Polyp	1000
m2caiSeg [113]	Endoscopy	Surgical tools and abdominal tissues	307
PolypGen [114] [115] [116]	Endoscopy	Polyp	1412
RoboTool [117]	Endoscopy	Surgical tools	500
sisvse [118]	Endoscopy	Surgical tools and abdominal tissues	18218
IDRiD* [119]	Fundus	Optic disc	81
PAPILA* [120]	Fundus	Optic disc and cup	488
REFUGE [121] [122]	Fundus	Optic disc and cup	1200
GlaS@MICCAI2015* [123] [124]	Pathology	Adenocarcinomas	165
HuBMAP HPA [125]	Pathology	Function tissue unit (FTU) of large intestine	58
HuBMAP Hacking the Kidney [126]	Pathology	Glomeruli FTU	8

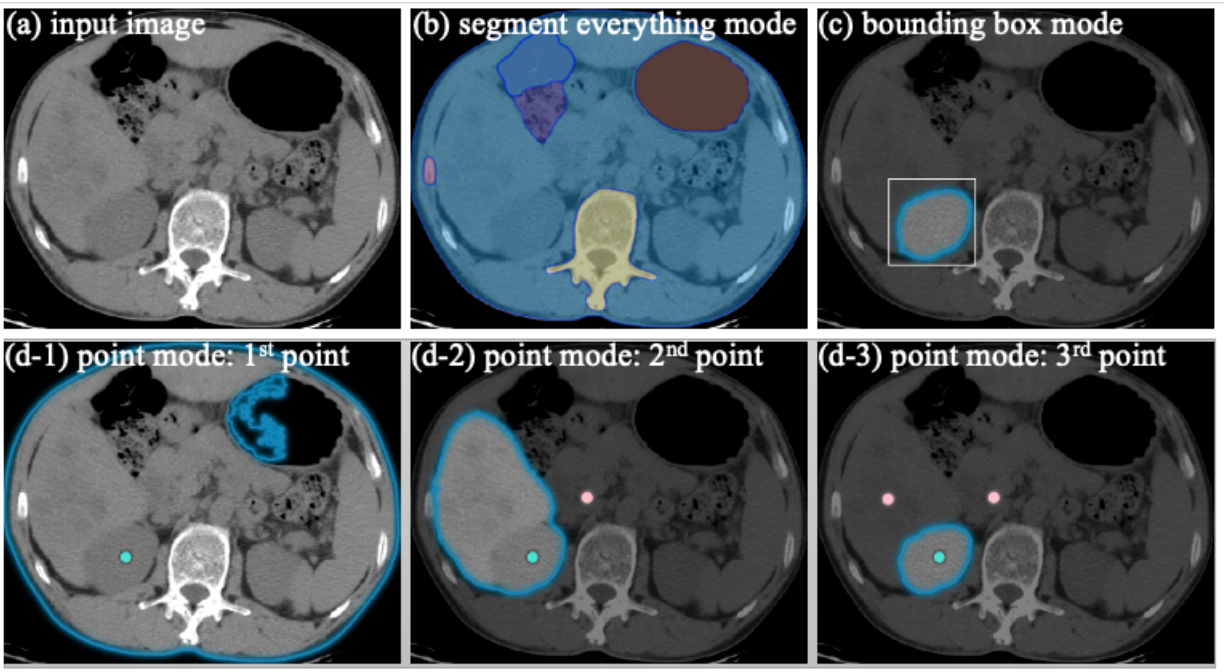

Fig. 1. Segmentation results of SAM based on different segmentation modes. The segment-everything mode divides the whole image into six regions based on the image intensity (b). However, such segmentation results have limited use because of two main reasons. On the one hand, the segmentation results do not have semantic labels. On the other hand, clinicians mainly focus on meaningful ROIs in clinical scenarios, e.g., the liver, kidneys, spleen, and lesions. The bounding box-based segmentation mode generates good results for the right kidney by just giving the upper-left and bottom-right points (c). For the point-based segmentation mode (d), we first give one foreground point to the center of the right kidney but the segmentation results include the whole abdomen tissues. Then, we add a background point on the over-segmentation regions. The segmentation mask shrinks to the liver and right kidney. After adding another background point on the liver, we finally obtain the expected kidney segmentation.

TABLE 5
Internal validation results of SAM, specialist U-Net, and MedSAM on CT image segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
Aora	CT	91.9 (89.3, 93.4)	93.1 (91.2, 94.9)*	95.6 (95.1, 96.1)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
COVID-19 Infection	CT	62.4 (45.5, 72.5)	71.3 (49.8, 81.7)*	84.2 (76.4, 90.1)*†	65.6 (48.8, 86.7)	67.2 (48.4, 89.5)	93.2 (74.2, 97.9)*†
Colon Cancer	CT	72.6 (64.5, 81.3)	80.3 (75.7, 83.9)*	83.0 (77.7, 86.4)*†	87.2 (86.5, 93.1)	93.7 (92.6, 95.2)*	97.4 (96.7, 98.1)*†
Esophagus	CT	71.5 (68.1, 75.5)	56.3 (51.4, 63.6)*	73.7 (68.8, 78.4)*†	91.4 (83.4, 94.4)	84.2 (55.6, 94.2)*	95.6 (86.8, 97.9)*†
Gallbladder	CT	85.4 (77.4, 90.1)	87.0 (82.8, 92.1)	91.8 (87.9, 95.2)*†	93.5 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Glioblastoma	CT	74.4 (69.8, 78.2)	86.2 (83.4, 88.4)*	94.3 (93.3, 95.4)*†	83.3 (81.9, 85.3)	89.4 (86.2, 91.5)*	99.4 (99.2, 99.6)*†
Head-Neck Cancer	CT	61.4 (44.2, 72.1)	67.3 (58.5, 75.1)*	79.4 (71.9, 83.8)*†	75.5 (69.1, 81.7)	70.7 (62.5, 80.1)*	90.5 (85.7, 94.8)*†
Heart	CT	91.5 (89.8, 93.8)	94.2 (93.4, 95.6)*	97.6 (97.4, 97.9)*†	94.0 (92.4, 95.0)	85.6 (83.2, 88.2)*	96.3 (95.3, 97.3)*†
Inferior Vena Cava	CT	82.9 (78.6, 86.1)	85.4 (82.1, 88.6)*	91.8 (88.9, 93.6)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Intracranial Hemorrhage	CT	86.7 (82.9, 91.6)	89.0 (83.4, 93.4)*	94.0 (91.5, 94.9)*†	94.9 (91.0, 98.6)	98.6 (96.6, 99.6)*	99.4 (99.1, 99.7)*†
Kidney Cancer	CT	86.7 (82.6, 90.3)	87.7 (80.9, 92.0)	90.2 (86.5, 93.7)*†	93.1 (90.9, 95.8)	93.3 (87.6, 96.6)	98.7 (97.5, 99.4)*†
Left Adrenal Gland	CT	60.0 (52.0, 69.0)	52.2 (49.6, 58.3)*	65.4 (59.9, 74.4)†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Left Kidney	CT	94.6 (92.5, 96.6)	95.5 (94.4, 96.6)	97.0 (96.6, 97.7)*†	93.5 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.3)*†
Left Lung	CT	94.4 (92.3, 95.6)	95.6 (93.8, 96.5)*	95.7 (95.0, 96.3)*†	94.2 (90.4, 96.6)	94.9 (92.2, 96.8)*	98.0 (96.4, 98.9)*†
Liver	CT	93.2 (91.6, 94.7)	96.6 (96.1, 97.1)*	98.0 (97.8, 98.4)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Liver Cancer	CT	80.0 (72.6, 83.8)	85.2 (80.1, 90.9)*	88.7 (84.2, 93.0)*†	90.1 (86.3, 93.3)	93.3 (89.6, 96.6)*	97.9 (96.2, 98.5)*†
Lung Cancer	CT	73.2 (64.2, 81.6)	85.5 (78.9, 88.3)*	94.1 (90.4, 95.5)*†	89.4 (84.4, 93.1)	91.9 (88.8, 94.7)*	99.5 (98.7, 99.7)*†
Lymph Nodes	CT	77.1 (74.1, 82.1)	69.2 (67.8, 77.3)*	82.1 (78.8, 87.6)*†	90.2 (85.7, 93.0)	79.6 (75.3, 84.1)*	93.4 (91.9, 96.4)*†
Melanoma	CT	63.7 (54.5, 70.6)	78.0 (73.3, 82.9)*	83.8 (79.6, 87.6)*†	80.4 (77.3, 83.7)	86.6 (83.9, 91.1)*	94.0 (92.3, 95.4)*†
Pancreas	CT	73.1 (66.9, 76.9)	80.3 (76.1, 82.1)*	87.2 (85.9, 90.5)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Pancreas Cancer	CT	74.1 (69.1, 77.2)	69.9 (62.1, 80.3)*	79.1 (75.5, 83.2)*†	84.7 (81.1, 88.6)	80.5 (76.2, 89.2)*	95.0 (92.1, 97.5)*†
Pleural Effusion	CT	35.3 (21.7, 53.7)	70.9 (62.5, 80.4)*	82.1 (71.6, 92.9)*†	61.3 (60.6, 71.5)	80.2 (77.9, 86.8)*	95.2 (86.9, 98.6)*†
Right Adrenal Gland	CT	51.5 (47.9, 59.6)	49.8 (44.9, 57.5)	66.8 (57.7, 72.6)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Right Kidney	CT	94.8 (92.3, 96.6)	95.8 (94.3, 96.8)*	97.1 (96.4, 97.9)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Right Lung	CT	95.2 (92.6, 96.0)	95.8 (94.1, 96.7)*	97.1 (96.6, 97.7)*†	94.2 (90.4, 96.6)	94.9 (92.2, 96.8)	98.0 (96.4, 98.9)*†
Spinal Cord	CT	75.4 (70.9, 78.9)	54.2 (51.4, 57.9)*	78.1 (75.1, 80.2)*†	87.5 (81.6, 94.0)	59.2 (53.8, 85.0)*	90.0 (84.7, 95.9)*†
Spleen	CT	94.3 (92.3, 96.5)	96.0 (95.1, 97.4)*	97.6 (97.2, 98.2)*†	93.5 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.3)*†
Stomach	CT	86.3 (78.3, 90.2)	92.8 (90.7, 94.3)*	96.2 (95.1, 97.2)*†	93.6 (91.8, 94.8)	97.2 (96.1, 97.9)*	99.0 (98.5, 99.4)*†
Throat Cancer	CT	28.1 (20.9, 53.6)	66.6 (66.4, 78.1)*	80.3 (79.3, 86.9)*†	74.4 (69.1, 92.1)	85.5 (80.7, 90.4)	98.1 (98.1, 99.5)*†

TABLE 6
Internal validation results of SAM, specialist U-Net, and MedSAM on MR image segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
Brainstem	MR	69.2 (65.4, 75.8)	89.9 (88.7, 91.1)*	97.1 (96.7, 97.2)*†	62.6 (57.3, 67.9)	93.8 (92.9, 97.5)*	100 (99.6, 100)*†
Cerebellum	MR	76.5 (72.1, 78.7)	91.0 (86.9, 91.5)*	96.8 (96.3, 97.2)*†	58.7 (52.1, 64.5)	91.8 (86.1, 93.5)*	99.8 (99.1, 99.9)*†
Deep Grey Matter	MR	49.6 (25.4, 60.0)	80.1 (74.4, 87.8)*	95.6 (92.7, 95.9)*†	50.4 (48.3, 54.0)	76.9 (69.1, 80.8)*	96.6 (94.4, 98.6)*†
Glioma	MR-FLAIR	83.4 (78.9, 87.0)	92.8 (90.2, 94.5)*	96.2 (94.4, 97.1)*†	70.5 (62.4, 78.6)	93.9 (90.7, 96.9)*	98.8 (97.0, 99.7)*†
Glioma	MR-T1	76.3 (70.6, 80.3)	87.8 (84.2, 90.5)*	94.4 (91.6, 95.8)*†	56.1 (50.7, 62.8)	82.3 (76.2, 87.7)*	97.6 (93.9, 99.0)*†
Glioma Enhancing Tumor	MR-T2	78.8 (73.8, 83.8)	86.9 (82.2, 90.4)*	95.2 (92.7, 96.4)*†	68.8 (60.8, 76.7)	84.7 (79.1, 88.9)*	99.0 (96.2, 99.8)*†
Glioma Tumor Core	MR-T1-CE	71.0 (62.5, 79.1)	90.5 (84.7, 94.2)*	95.9 (92.7, 97.3)*†	74.4 (67.1, 83.8)	97.9 (92.8, 99.5)*	99.8 (98.9, 100)*†
Ischemic Stroke	MR-ADC	61.3 (54.4, 68.9)	82.9 (75.2, 87.5)*	92.3 (87.3, 95.3)*†	75.9 (69.9, 81.0)	95.6 (90.4, 98.7)*	94.5 (91.2, 98.0)*
Left Atrium	MR	83.6 (82.8, 83.7)	90.8 (90.3, 91.3)	97.3 (97.1, 97.6)	70.2 (67.3, 73.2)	90.6 (87.8, 92.7)	99.1 (98.4, 99.4)
Left Ventricle	MR	77.5 (76.1, 81.8)	94.5 (93.6, 95.7)*	98.5 (97.9, 98.7)*†	85.1 (79.2, 88.8)	99.6 (99.3, 99.8)*	100 (100, 100)*†
Meningioma	MR-T1-CE	92.1 (87.4, 93.5)	94.6 (90.3, 95.2)*	97.9 (97.6, 98.1)*†	95.5 (92.6, 97.6)	98.6 (96.7, 99.7)*	100 (100, 100)*†
Meningiomai	MR-T2-FLAIR	79.2 (72.8, 85.4)	89.2 (88.7, 92.1)*	97.0 (96.2, 97.6)*†	66.4 (61.3, 83.5)	94.7 (90.6, 98.6)*	100 (99.4, 100)*†
Prostate	MR-ADC	83.7 (81.0, 85.1)	91.1 (89.7, 92.6)*	98.5 (98.2, 98.5)*†	83.2 (82.0, 85.4)	96.0 (93.5, 97.4)*	100 (100, 100)*†
Prostate Cancer	MR	69.3 (64.4, 72.2)	79.2 (74.8, 81.8)*	96.9 (93.1, 97.5)*†	78.1 (73.6, 81.2)	89.8 (84.6, 94.0)*	100 (99.3, 100)*†
Prostate	MR-T2	90.7 (89.0, 91.5)	93.8 (93.0, 94.4)*	98.4 (98.3, 98.6)*†	90.1 (88.0, 92.7)	97.2 (95.3, 98.0)*	100 (100, 100)*†
Right Ventricle	MR	90.3 (88.1, 92.5)	93.9 (91.7, 95.2)*	97.2 (96.7, 98.1)*†	95.6 (91.1, 98.6)	99.7 (99.3, 100)*	100 (100, 100)*†
Spine	MR	80.8 (77.4, 84.0)	87.7 (85.6, 88.6)*	91.8 (90.9, 93.1)*†	86.0 (83.5, 89.2)	92.9 (90.9, 93.7)*	96.8 (96.1, 97.6)*†
Ventricles	MR	63.9 (47.9, 69.8)	69.5 (56.0, 73.4)*	90.0 (79.4, 91.9)*†	63.0 (56.2, 66.4)	66.8 (61.8, 68.8)*	85.1 (78.7, 90.7)*†
Vestibular Schwannoma	MR	85.3 (78.5, 89.5)	90.3 (80.3, 93.2)*	95.2 (91.6, 97.7)*†	97.0 (94.5, 97.9)	99.2 (96.7, 99.9)*	99.9 (98.4, 100)*†

TABLE 7
Internal validation results of SAM, specialist U-Net, and MedSAM on grey images (Chest X-Ray (CXR), Mammography, OCT, and Ultrasound) segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
COVID-19	CXR	78.2 (70.6, 85.3)	91.1 (86.5, 94.7)*	92.9 (89.3, 95.6)*†	86.5 (80.8, 92.0)	97.2 (94.5, 98.9)*	98.3 (96.5, 99.3)*†
Heart	CXR	90.1 (80.5, 93.5)	95.6 (94.5, 96.2)*	96.8 (96.0, 97.3)*†	94.7 (85.9, 97.5)	98.7 (98.2, 99.2)*	99.4 (98.8, 99.7)*†
Lung	CXR	93.3 (90.1, 95.2)	98.2 (97.5, 98.5)*	99.1 (98.3, 99.4)*†	97.6 (94.6, 99.0)	100 (99.8, 100)*	100 (99.9, 100)*†
Pneumothorax	CXR	50.2 (38.1, 62.9)	76.3 (67.8, 83.2)*	81.5 (75.1, 86.8)*†	67.5 (54.3, 81.0)	92.6 (85.0, 96.7)*	96.4 (93.0, 98.2)*†
Tuberculosis	CXR	93.9 (89.7, 95.6)	96.8 (95.3, 97.4)*	96.9 (95.4, 97.6)*†	97.8 (93.7, 99.0)	99.7 (98.7, 99.8)*	99.6 (98.3, 99.9)*
Breast Cancer	Mammography	66.5 (46.5, 77.7)	78.4 (70.9, 85.4)*	83.3 (78.3, 87.5)*†	80.4 (60.6, 89.8)	91.8 (84.1, 96.0)*	94.8 (91.0, 97.3)*†
Diabetic Macular Edema	OCT	88.4 (83.1, 91.6)	91.5 (86.8, 95.1)*	95.0 (92.9, 96.3)*†	99.0 (97.1, 100)	99.9 (98.8, 100)*	100 (99.8, 100)*†
Benign Breast Cancer	Ultrasound	89.3 (81.9, 92.9)	93.5 (90.1, 94.8)*	94.0 (92.6, 95.2)*†	96.4 (88.6, 99.2)	98.8 (97.6, 99.5)*	99.1 (98.0, 99.7)*
Benign Thyroid Nodule	Ultrasound	88.1 (83.0, 91.9)	93.8 (91.1, 95.8)*	94.8 (92.8, 96.3)*†	93.7 (89.5, 96.6)	98.0 (96.3, 99.1)*	98.7 (97.8, 99.4)*†
Fetal Head	Ultrasound	84.9 (78.7, 88.9)	96.1 (95.0, 96.9)*	98.1 (97.3, 98.6)*†	91.2 (86.3, 94.7)	99.4 (98.6, 99.8)*	100 (99.8, 100)*†
Kidney	Ultrasound	88.0 (82.6, 92.5)	96.9 (95.4, 97.7)*	98.1 (97.3, 98.6)*†	92.5 (87.5, 96.5)	99.6 (98.8, 99.8)*	99.9 (99.7, 100)*†
Left Atrium	Ultrasound	44.7 (4.2, 81.6)	94.1 (92.3, 95.4)*	98.3 (97.7, 98.7)*†	68.3 (7.4, 92.5)	98.9 (98.1, 99.5)*	100 (100, 100)*†
Left Ventricle	Ultrasound	81.8 (73.0, 86.3)	95.2 (93.8, 96.2)*	98.4 (97.9, 98.6)*†	91.0 (85.8, 93.9)	99.2 (98.6, 99.6)*	100 (100, 100)*†
Malignant Breast Cancer	Ultrasound	82.9 (72.0, 87.6)	88.2 (85.6, 91.0)*	91.3 (88.2, 92.9)*†	91.4 (80.1, 92.4)	93.7 (92.0, 96.2)*	95.8 (93.9, 97.4)*†
Malignant Thyroid Nodule	Ultrasound	88.1 (82.1, 91.8)	91.3 (88.4, 93.7)*	93.1 (90.7, 94.7)*†	95.6 (91.4, 97.9)	98.1 (96.5, 99.2)*	98.9 (97.7, 99.5)*†
Nerve Cancer	Ultrasound	55.3 (40.1, 71.0)	89.5 (87.0, 91.8)*	90.6 (88.2, 92.3)*†	75.0 (58.8, 85.4)	97.2 (95.2, 98.6)*	97.7 (96.2, 98.7)*†
Pubic Symphysis	Ultrasound	75.7 (67.3, 80.2)	92.6 (90.4, 94.5)*	97.1 (95.6, 98.3)*†	87.4 (81.8, 90.9)	99.6 (98.7, 99.9)*	100 (100, 100)*†

TABLE 8
Internal validation results of SAM, specialist U-Net, and MedSAM on RGB images (Dermoscopy, Endoscopy, Fundus, and Pathology) segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
Skin Cancer	Dermoscopy	88.8 (79.9, 93.2)	95.1 (92.2, 97.0)*	95.2 (93.0, 96.7)*†	94.9 (85.9, 98.3)	99.4 (98.6, 99.8)*	99.1 (97.7, 99.7)*†
Cholecystectomy	Endoscopy	71.8 (62.7, 81.5)	93.9 (89.9, 96.0)*	95.5 (93.3, 96.9)*†	81.1 (72.1, 89.8)	98.8 (96.3, 99.5)*	99.5 (98.7, 99.8)*†
Gastrectomy	Endoscopy	81.5 (72.4, 89.0)	89.2 (82.6, 94.4)*	97.4 (95.5, 98.7)*†	90.6 (82.4, 96.7)	96.5 (91.1, 99.1)*	99.9 (99.5, 100)*†
Polyp	Endoscopy	94.1 (89.9, 96.3)	94.8 (91.4, 96.2)	98.4 (97.9, 98.9)*†	99.1 (96.8, 99.8)	99.6 (98.2, 99.9)	100 (100, 100)*†
Surgical Instrument	Endoscopy	87.0 (80.3, 90.1)	96.5 (94.9, 97.3)*	96.7 (95.6, 97.5)*	93.9 (87.7, 96.9)	99.8 (99.4, 99.9)*	99.8 (99.5, 99.9)*
Glaucoma Cup	Fundus	79.0 (73.2, 83.2)	80.2 (75.5, 86.5)	97.8 (96.5, 98.3)*†	90.2 (87.4, 94.6)	95.6 (90.1, 96.5)*	100 (100, 100)*†
Glaucoma Disc	Fundus	85.4 (82.3, 87.2)	95.7 (94.1, 96.4)*	98.3 (97.5, 98.6)*†	96.8 (93.6, 97.5)	100 (99.6, 100)*	100 (100, 100)*†
Non-Glaucoma Cup	Fundus	69.8 (63.4, 75.6)	81.3 (74.2, 88.0)*	96.2 (95.4, 97.3)*†	90.1 (85.1, 95.6)	97.0 (90.2, 99.8)*	100 (100, 100)*†
Non-Glaucoma Disc	Fundus	85.1 (76.0, 89.3)	94.4 (93.3, 95.4)*	98.3 (98.0, 98.8)*†	95.7 (90.7, 98.6)	100 (99.8, 100)*	100 (100, 100)*†
Glomeruli	Pathology	93.3 (90.8, 95.2)	97.1 (96.2, 97.8)*	98.5 (98.3, 98.7)*†	98.1 (96.5, 99.2)	99.8 (99.6, 100)*	100 (100, 100)*†
Intestine FTU	Pathology	89.0 (84.6, 92.7)	93.9 (92.5, 95.1)*	94.2 (92.5, 95.0)*	96.7 (93.2, 98.4)	99.2 (98.4, 99.5)*	99.1 (98.6, 99.5)*

TABLE 9
External validation results of SAM, specialist U-Net, and MedSAM on CT image segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
Aora	CT	91.2 (89.7, 92.6)	85.3 (78.0, 89.4)*	93.1 (91.2, 94.7)*†	92.4 (89.9, 94.3)	81.6 (69.8, 90.7)*	98.7 (95.0, 99.6)*†
Esophagus	CT	68.0 (62.3, 73.9)	48.9 (41.5, 58.5)*	67.4 (59.6, 77.2)†	81.8 (76.0, 87.1)	47.2 (37.3, 60.7)*	80.3 (71.9, 89.7)†
Gallbladder	CT	78.9 (71.2, 86.0)	72.2 (62.1, 80.4)*	84.6 (76.7, 88.6)*†	88.6 (83.2, 93.3)	78.0 (70.2, 86.6)*	93.9 (87.5, 98.0)*†
Heart	CT	91.3 (90.2, 92.0)	90.7 (88.6, 91.9)	96.8 (96.7, 97.2)*†	88.8 (86.9, 90.9)	88.5 (84.0, 89.6)	99.5 (99.3, 99.8)*†
Inferior Vena Cava	CT	76.2 (71.0, 79.5)	63.0 (57.2, 70.6)*	79.5 (71.6, 84.0)*†	72.2 (67.9, 77.5)	52.6 (46.9, 61.2)*	78.8 (69.5, 86.6)*†
Left Kidney	CT	93.7 (92.6, 94.7)	93.0 (92.3, 93.8)*	93.7 (92.9, 94.4)†	92.7 (90.6, 95.8)	94.1 (92.2, 96.2)*	94.9 (93.3, 96.8)*†
Left Lung	CT	91.5 (89.6, 94.3)	92.4 (90.8, 94.3)	96.6 (96.1, 97.2)*†	86.9 (83.4, 92.9)	88.9 (84.1, 92.7)	98.9 (98.1, 99.5)*†
Liver	CT	91.6 (88.4, 93.4)	94.3 (93.1, 95.0)*	94.8 (93.6, 95.6)*†	76.4 (70.4, 82.3)	90.2 (86.6, 92.7)*	93.4 (89.5, 95.5)*†
Liver Cancer	CT	76.6 (70.1, 85.1)	87.3 (82.4, 91.5)*	96.0 (94.7, 96.9)*†	67.7 (59.3, 77.1)	80.0 (75.6, 84.3)*	98.6 (97.1, 99.5)*†
Lung Cancer	CT	64.2 (53.5, 70.9)	55.8 (52.0, 70.3)*	77.8 (68.7, 90.2)*†	77.5 (74.4, 83.7)	68.1 (63.1, 79.4)*	89.5 (86.5, 98.0)*†
Nasopharynx Cancer	CT	37.0 (31.5, 44.4)	65.8 (61.5, 70.9)*	90.3 (87.8, 93.2)*†	71.0 (66.1, 75.5)	68.7 (65.5, 75.4)	97.9 (96.6, 99.7)*†
Pancreas	CT	72.8 (65.9, 77.0)	73.7 (69.5, 78.1)*	80.6 (77.4, 84.9)*†	77.0 (71.7, 81.3)	81.6 (75.4, 86.5)*	90.0 (85.0, 93.5)*†
Right Kidney	CT	93.5 (92.9, 94.4)	93.2 (92.4, 94.0)*	93.5 (93.0, 94.2)†	92.2 (90.4, 94.3)	94.3 (92.1, 96.0)*	94.2 (92.6, 95.9)*
Right Lung	CT	92.6 (91.3, 94.5)	93.6 (92.1, 94.9)	97.4 (97.0, 97.7)*†	83.3 (80.6, 90.8)	89.9 (83.2, 93.0)*	99.2 (98.8, 99.6)*†
Spinal Cord	CT	47.5 (39.6, 56.2)	38.6 (35.5, 41.8)*	58.1 (55.8, 62.4)*†	58.5 (53.2, 63.9)	26.6 (21.6, 31.5)*	58.9 (50.3, 65.1)†
Spleen	CT	93.3 (89.6, 95.1)	94.0 (93.0, 94.8)*	94.5 (93.6, 95.2)*†	92.2 (84.5, 96.3)	96.8 (94.7, 98.2)*	98.7 (97.8, 99.2)*†
Stomach	CT	85.5 (80.5, 89.6)	90.1 (87.2, 92.3)*	93.1 (91.4, 94.5)*†	77.8 (71.6, 82.5)	85.8 (81.1, 91.2)*	94.8 (90.7, 97.5)*†

TABLE 10
External validation results of SAM, specialist U-Net, and MedSAM on MR image segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
A Carotid L	MR	57.8 (54.7, 62.3)	50.7 (47.9, 53.7)*	62.0 (59.5, 65.4)*†	70.0 (65.6, 73.1)	58.7 (53.3, 63.4)*	74.5 (68.8, 79.1)*†
A Carotid R	MR	61.0 (55.6, 64.2)	51.5 (48.6, 55.2)*	62.7 (59.3, 66.9)*†	72.0 (68.2, 75.1)	58.3 (56.4, 61.2)*	74.1 (69.3, 81.2)*†
Bone Mandible	MR	54.3 (50.2, 58.3)	22.6 (19.5, 25.9)*	69.7 (66.7, 73.7)*†	47.9 (45.7, 51.9)	35.1 (31.8, 38.6)*	60.0 (56.7, 63.9)*†
Brain Cancer	MR	85.4 (78.1, 90.4)	82.5 (76.7, 88.1)	94.5 (93.6, 96.1)*†	87.3 (75.7, 96.4)	91.6 (81.3, 94.2)	100 (99.8, 100)*†
Brainstem	MR	84.4 (83.3, 86.9)	88.6 (88.0, 89.3)*	96.0 (95.6, 96.2)*†	79.8 (77.6, 85.5)	88.1 (85.6, 89.7)*	99.7 (99.5, 99.9)*†
Cervical Cancer	MR	76.9 (70.1, 81.6)	81.6 (78.0, 86.0)*	89.7 (88.3, 90.9)*†	88.6 (84.9, 90.5)	92.8 (91.4, 94.2)	98.2 (97.2, 99.1)*†
Cricopharyngeus	MR	61.4 (55.5, 64.6)	61.1 (56.6, 64.0)	90.2 (83.1, 91.5)*†	68.5 (63.1, 73.3)	71.5 (68.8, 79.6)*	99.2 (97.0, 99.8)*†
Eye PL	MR	81.5 (78.5, 83.6)	82.7 (81.1, 83.8)*	94.1 (93.3, 94.7)*†	84.1 (78.4, 88.5)	86.6 (82.7, 89.9)*	99.6 (99.1, 99.8)*†
Eye PR	MR	81.9 (80.1, 83.8)	81.9 (79.4, 83.8)	94.0 (93.0, 94.4)*†	85.8 (80.5, 91.8)	85.7 (83.1, 88.2)	99.6 (99.3, 99.9)*†
Glnd Lacrimal L	MR	61.3 (58.6, 63.4)	51.5 (50.1, 56.2)*	64.0 (62.5, 66.7)*†	74.1 (70.7, 80.7)	59.3 (55.6, 66.9)*	79.7 (74.3, 86.1)*†
Glnd Lacrimal R	MR	59.9 (57.5, 66.9)	51.6 (48.7, 60.1)*	68.7 (60.5, 76.0)*†	79.8 (75.0, 84.3)	65.1 (58.3, 69.0)*	88.9 (79.7, 92.6)*†
Glnd Submand L	MR	77.9 (74.9, 81.1)	76.6 (74.3, 79.1)	91.3 (90.2, 91.9)*†	81.4 (76.9, 85.4)	80.2 (73.7, 83.3)*	98.7 (98.0, 99.3)*†
Glnd Submand R	MR	79.7 (74.3, 82.7)	77.2 (72.9, 79.6)*	90.9 (89.2, 92.5)*†	85.8 (78.7, 88.6)	81.0 (74.1, 83.8)*	98.9 (98.4, 99.4)*†
Glottis	MR	30.1 (14.4, 42.7)	44.2 (35.6, 49.6)*	85.0 (79.1, 88.6)*†	61.6 (53.7, 67.7)	66.6 (62.0, 71.1)*	97.9 (95.6, 99.5)*†
Heart	MR	52.1 (36.3, 66.8)	91.3 (88.5, 93.1)*	96.3 (95.2, 96.9)*†	75.0 (62.8, 84.5)	99.2 (98.2, 99.7)*	100 (99.9, 100)*†
Larynx SG	MR	54.0 (46.9, 60.6)	60.4 (56.3, 64.5)*	88.2 (85.2, 90.4)*†	71.0 (65.1, 75.3)	72.4 (65.6, 78.2)	97.4 (95.9, 98.8)*†
Left Kidney	T1-Inphase	89.2 (86.0, 92.1)	91.8 (88.5, 92.3)	95.0 (93.4, 96.5)*†	94.8 (92.0, 97.8)	96.6 (94.3, 97.7)	99.9 (99.6, 100)*†
Left Kidney	T1-Outphase	89.7 (86.6, 91.9)	89.6 (88.0, 91.0)	93.3 (92.9, 94.3)*†	96.0 (92.4, 98.3)	96.0 (94.2, 97.3)	99.6 (99.3, 99.9)*†
Left Kidney	T2	92.8 (91.2, 94.3)	94.1 (93.2, 94.8)*	96.1 (94.7, 96.7)*†	97.6 (94.3, 98.3)	98.4 (97.3, 99.3)*	99.6 (99.4, 99.9)*†
Lips	MR	58.4 (50.9, 66.5)	60.3 (50.6, 67.1)	86.9 (83.9, 89.3)*†	69.2 (63.6, 72.5)	62.6 (59.3, 68.2)*	96.3 (92.8, 96.8)*†
Liver	T1-Inphase	89.0 (88.0, 90.0)	92.1 (90.0, 92.8)*	95.6 (95.3, 96.0)*†	89.5 (82.9, 91.4)	92.2 (90.9, 95.4)*	98.7 (98.2, 99.2)*†
Liver	T1-Outphase	90.9 (89.7, 92.6)	93.6 (92.3, 94.0)*	95.0 (94.5, 96.1)*†	92.9 (88.4, 95.4)	96.2 (94.7, 97.4)*	98.7 (97.5, 99.1)*†
Liver	T2	90.4 (89.0, 91.7)	92.4 (91.3, 93.2)*	96.4 (95.2, 96.7)*†	91.1 (88.6, 93.2)	94.6 (91.7, 96.1)*	98.9 (98.4, 99.7)*†
OpticNrv L	MR	39.5 (34.2, 50.0)	41.9 (38.3, 54.3)*	61.3 (59.6, 66.3)*†	56.3 (52.3, 72.4)	55.5 (51.7, 70.6)	83.9 (79.2, 87.1)*†
OpticNrv R	MR	43.3 (29.7, 49.6)	41.3 (35.0, 47.1)	70.3 (64.6, 77.5)*†	59.4 (47.2, 66.3)	53.2 (47.4, 59.0)	92.4 (85.3, 94.0)*†
Parotid L	MR	72.7 (63.5, 77.5)	79.1 (77.5, 81.6)*	91.7 (90.6, 92.4)*†	67.5 (59.8, 75.0)	74.7 (68.6, 79.8)*	96.8 (95.4, 97.6)*†
Parotid R	MR	71.4 (67.5, 75.3)	77.3 (75.0, 79.3)*	91.6 (90.5, 92.3)*†	69.4 (63.3, 74.6)	69.4 (65.5, 75.0)	96.8 (95.8, 97.7)*†
Right Kidney	T1-Inphase	91.0 (87.6, 93.0)	91.0 (88.4, 92.6)	94.8 (94.1, 96.5)*†	97.2 (94.8, 98.9)	96.3 (94.1, 98.2)	99.9 (99.8, 100)*†
Right Kidney	T1-Outphase	89.4 (85.7, 91.4)	89.0 (87.5, 90.6)	93.6 (92.5, 94.9)*†	94.3 (92.6, 97.9)	95.2 (92.4, 96.9)	99.8 (99.5, 99.9)*†
Right Kidney	T2	94.0 (93.2, 94.4)	93.9 (92.2, 95.5)	95.9 (94.6, 97.1)*†	98.6 (97.7, 99.0)	97.1 (95.4, 98.0)*	99.9 (99.0, 100)*†
SpinalCord	MR	55.9 (54.3, 58.7)	64.3 (60.7, 67.7)*	77.4 (75.3, 80.6)*†	40.8 (35.9, 45.0)	57.9 (52.6, 63.0)*	85.5 (79.4, 91.1)*†
Spleen	T1-Inphase	86.4 (83.7, 88.5)	87.8 (86.1, 89.7)	94.3 (92.9, 95.0)*†	91.5 (85.3, 95.7)	93.6 (89.8, 94.7)	99.9 (99.7, 99.9)*†
Spleen	T1-Outphase	88.8 (86.3, 90.4)	88.6 (86.8, 91.4)	93.0 (91.6, 93.9)*†	96.5 (93.4, 98.8)	96.0 (88.8, 98.6)	99.7 (99.4, 99.9)*†
Spleen	T2	94.4 (92.9, 95.1)	91.9 (90.1, 94.7)	97.0 (95.5, 97.2)*†	99.7 (97.8, 99.9)	94.2 (89.7, 99.2)*	100 (99.9, 100)*†

TABLE 11
External validation results of SAM, specialist U-Net, and MedSAM on Chest X-Ray (CXR), Dermoscopy, Endoscopy, Fundus, and Pathology image segmentation tasks in terms of DSC and NSD. Scores are displayed as Median values (First quartile, Third quartile). Scores marked with $*$ indicate statistically significant superiority over SAM, while those marked with † denote statistically significant superiority of MedSAM over the specialist U-Net model (p-value < 0.05).

Target	Modality	DSC (%)			NSD (%)		
		SAM	U-Net	MedSAM	SAM	U-Net	MedSAM
COVID-19	CXR	91.4 (88.2, 94.1)	97.7 (97.1, 98.2)*	98.9 (98.3, 99.2)*†	96.1 (93.3, 98.3)	99.9 (99.8, 100)*	100 (99.9, 100)*†
Lung Opacity	CXR	89.0 (84.7, 92.0)	97.2 (96.1, 97.8)*	98.3 (97.6, 98.8)*†	94.6 (90.9, 97.0)	99.8 (99.4, 99.9)*	99.9 (99.7, 100)*†
Viral Pneumonia	CXR	89.2 (82.7, 92.3)	96.8 (95.6, 97.3)*	98.4 (97.5, 98.8)*†	94.9 (88.9, 97.2)	99.8 (99.3, 99.9)*	99.9 (99.7, 100)*†
Skin Cancer	Dermoscopy	94.1 (91.6, 96.0)	90.5 (86.7, 95.5)*	96.9 (96.3, 97.4)*†	99.4 (96.6, 99.8)	97.7 (95.9, 99.4)	99.9 (99.6, 100)*†
Polyp	Endoscopy	93.8 (89.8, 95.8)	95.9 (93.9, 97.0)*	98.0 (97.6, 98.4)*†	98.2 (95.8, 99.4)	99.3 (98.1, 99.7)*	99.9 (99.8, 100)*†
Optic Disc	Fundus	82.8 (73.4, 89.3)	93.0 (85.1, 95.5)*	97.4 (96.0, 98.1)*†	95.1 (89.1, 98.2)	99.2 (97.0, 99.8)*	100 (100, 100)*†
Colon Gland	Pathology	81.3 (75.4, 84.8)	96.0 (95.2, 96.7)*	95.9 (95.4, 96.7)*	89.1 (85.4, 92.1)	99.6 (99.2, 99.8)*	99.6 (99.3, 99.8)*

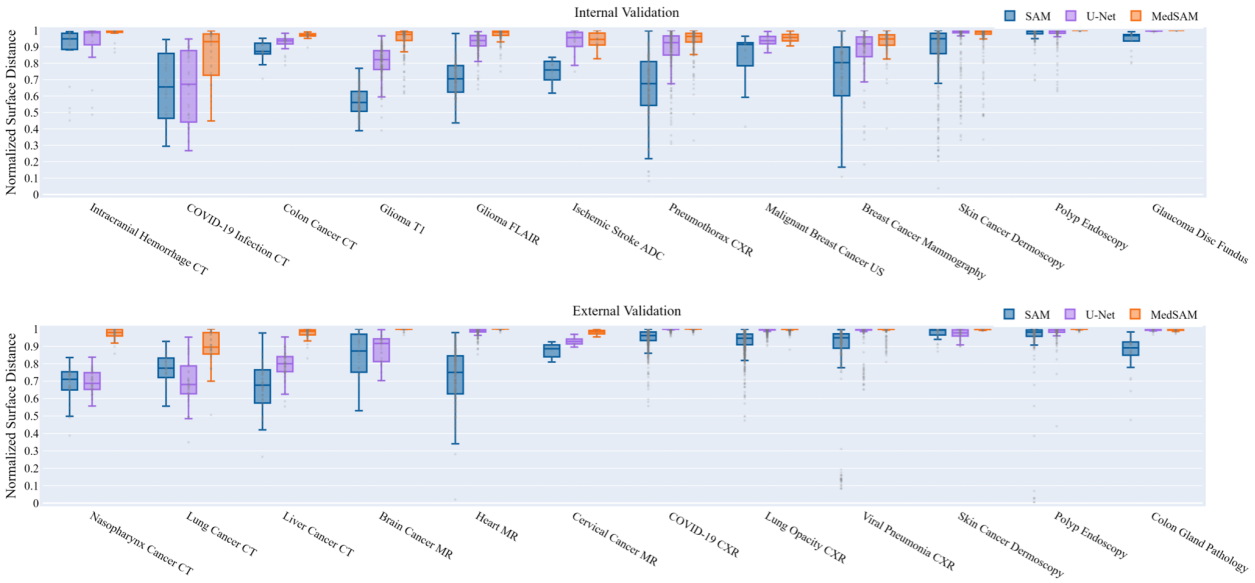

Fig. 2. Normalized Surface Distance for 12 representative segmentation tasks as part of the internal validation, and a further 12 tasks as part of the external validation (corresponding to tasks presented in Fig. 2 of the main text).

Fig. 3. Dice Similarity Coefficient of segmentation tasks in internal validation.

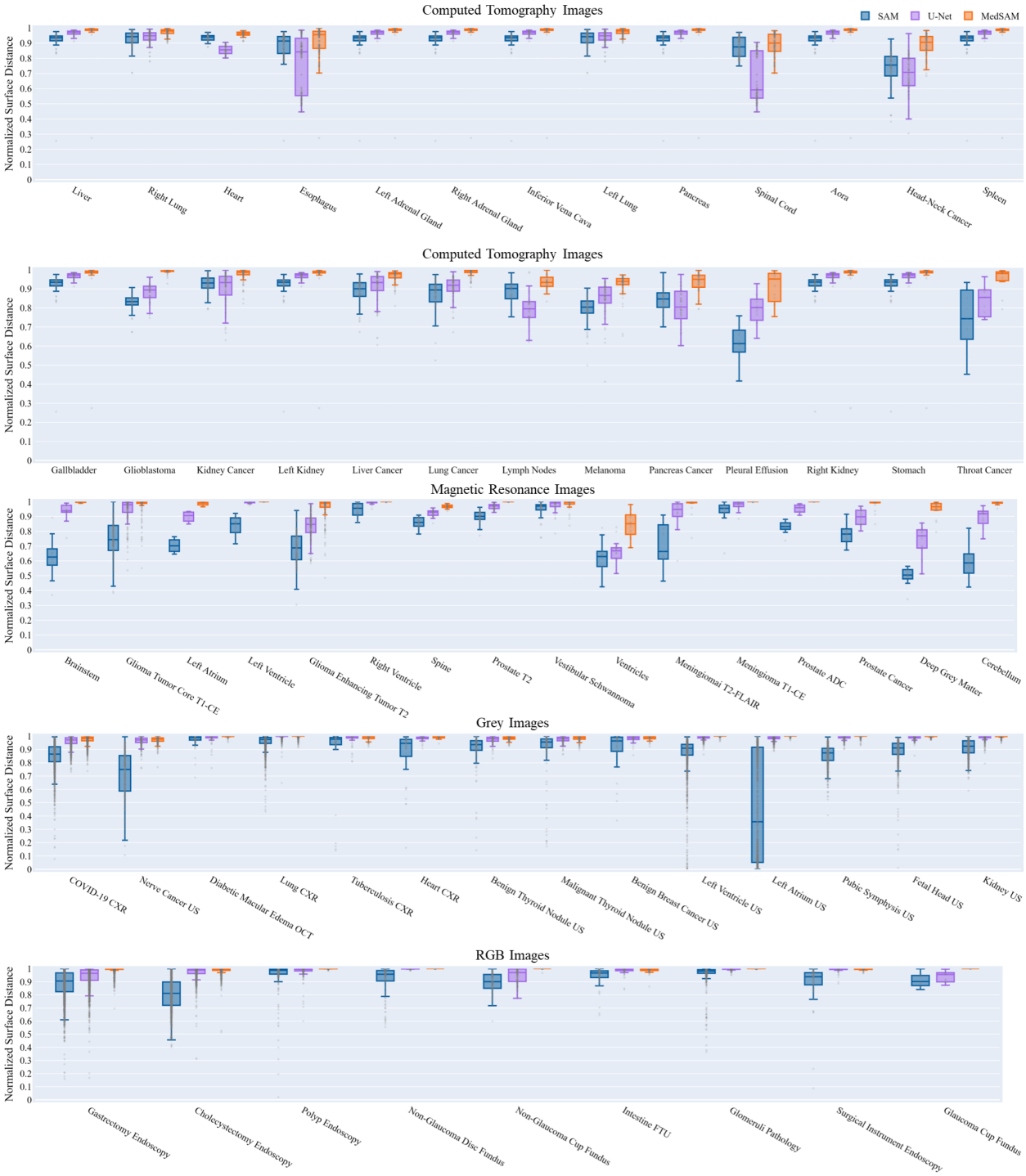

Fig. 4. Normalized surface distance of segmentation tasks in internal validation.

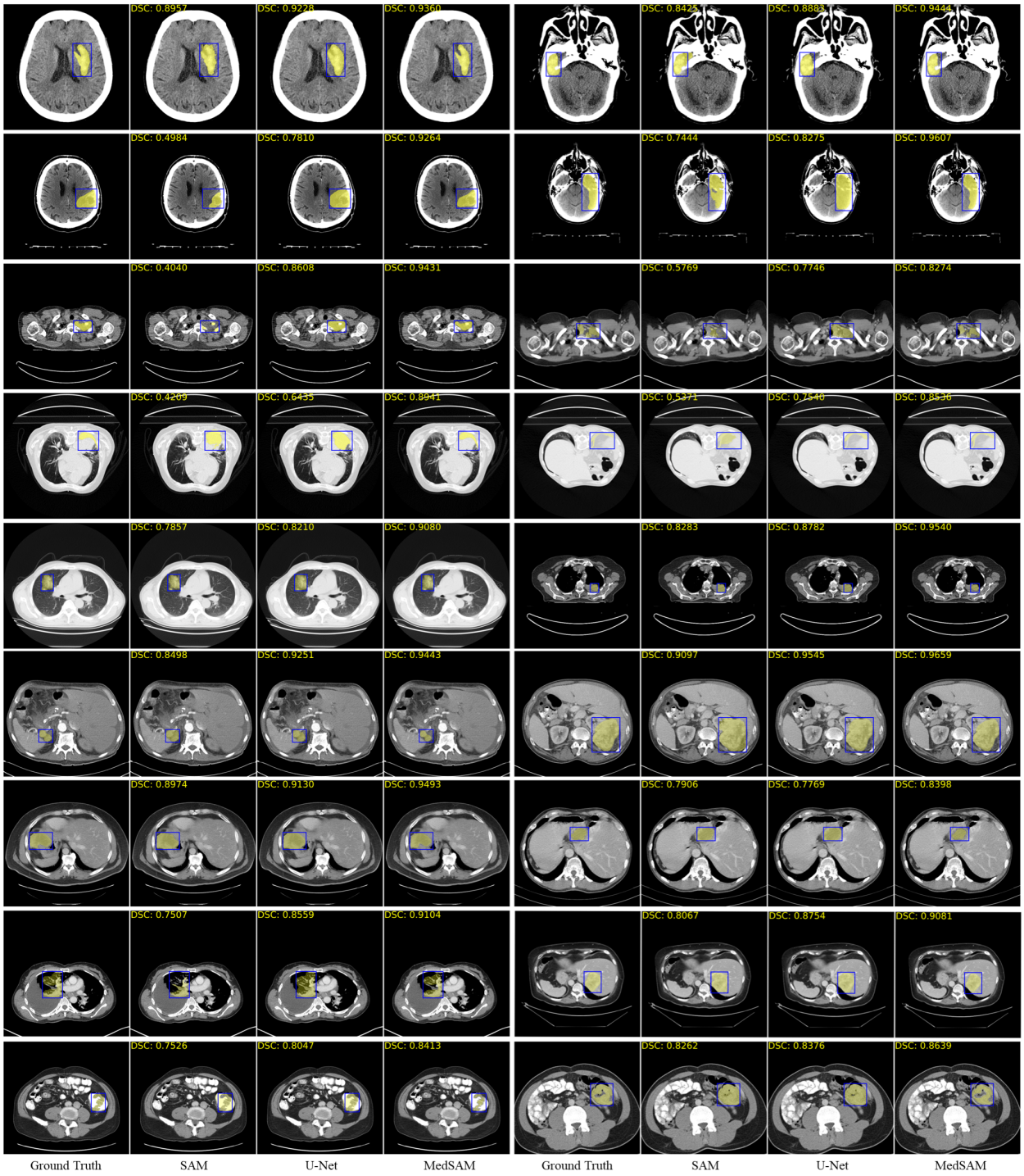

Fig. 5. Segmentation examples of CT images in internal validation.

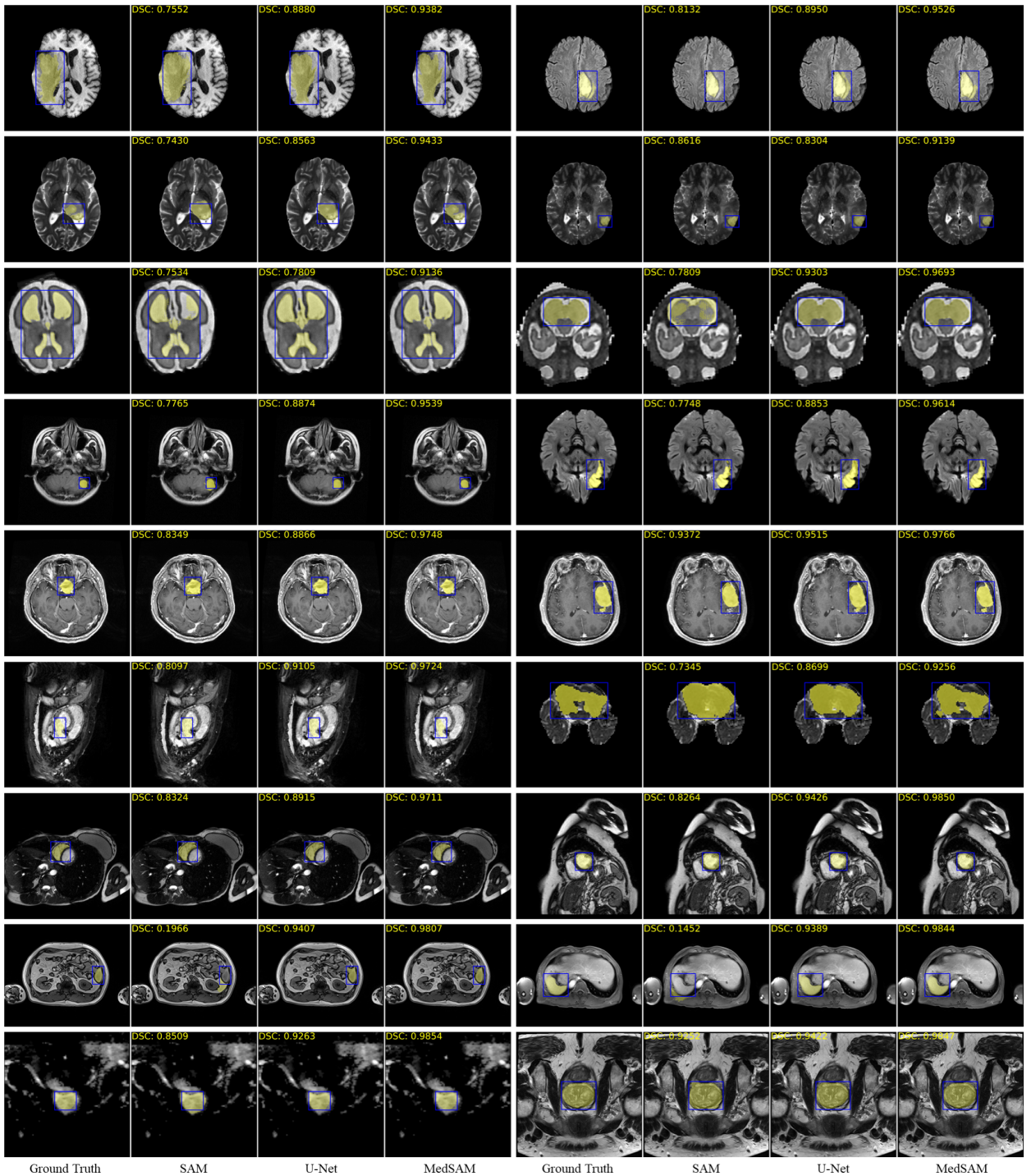

Fig. 6. Segmentation examples of MR images in internal validation.

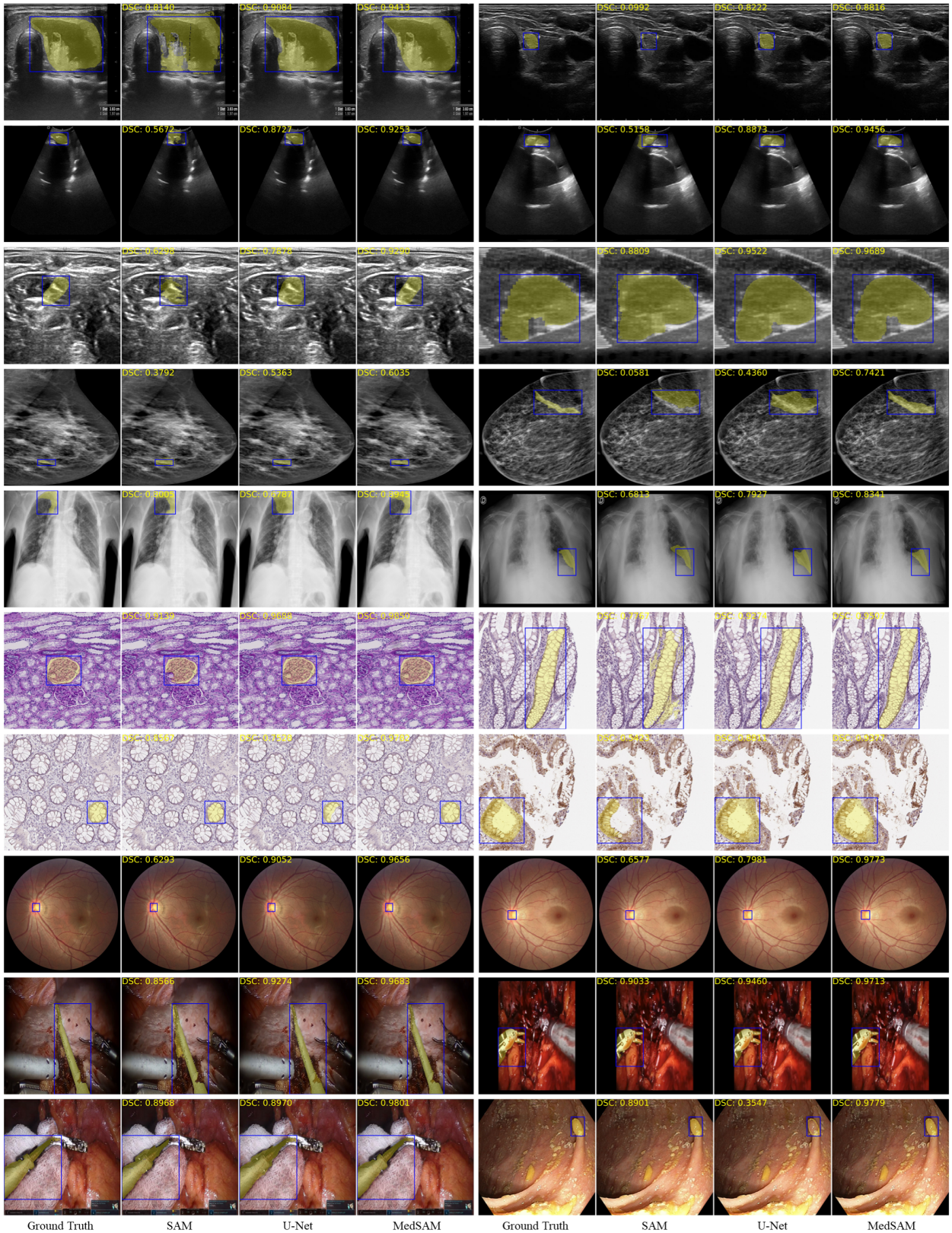

Fig. 7. Segmentation example of ultrasound, chest X-ray, pathology, fundus, and endoscopy images in internal validation.

Fig. 8. Dice similarity coefficient of segmentation tasks in external validation.

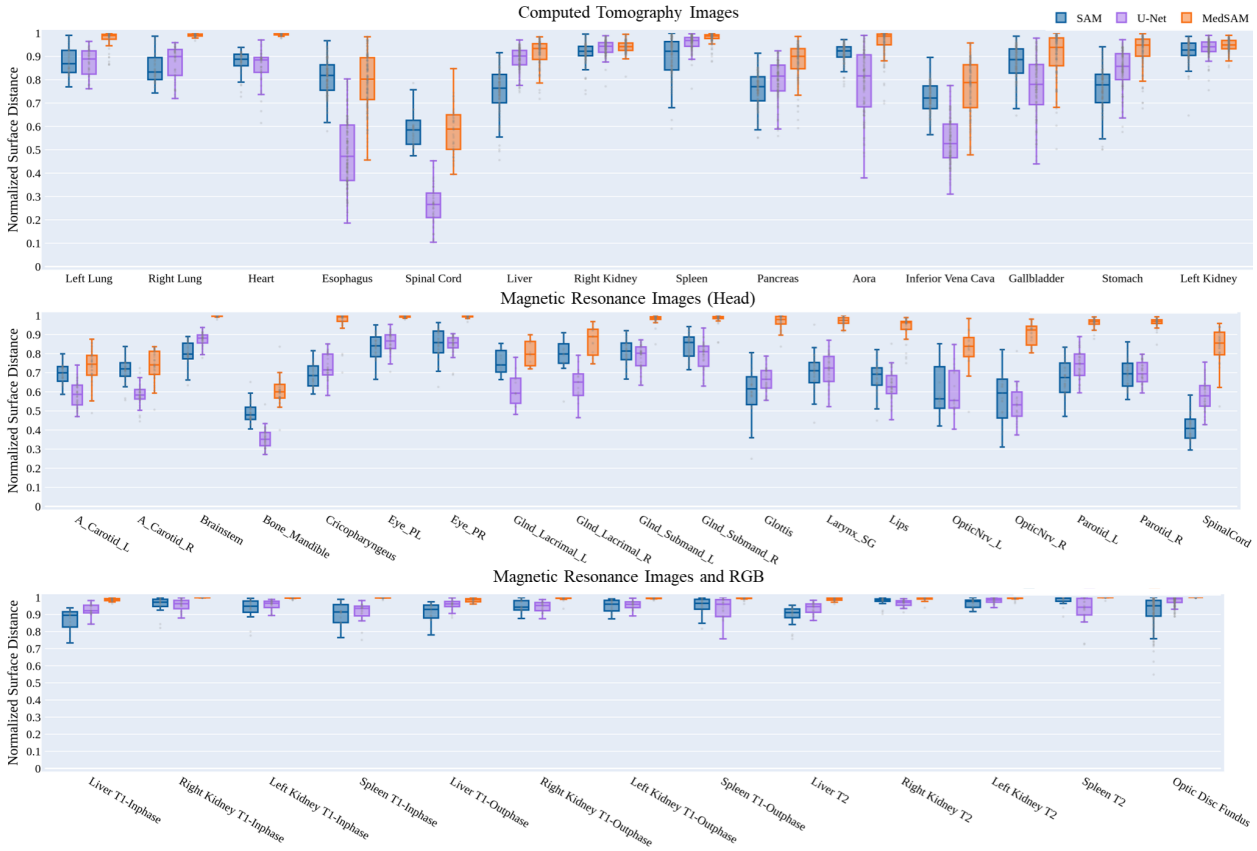

Fig. 9. Normalized surface distance of segmentation tasks in external validation.

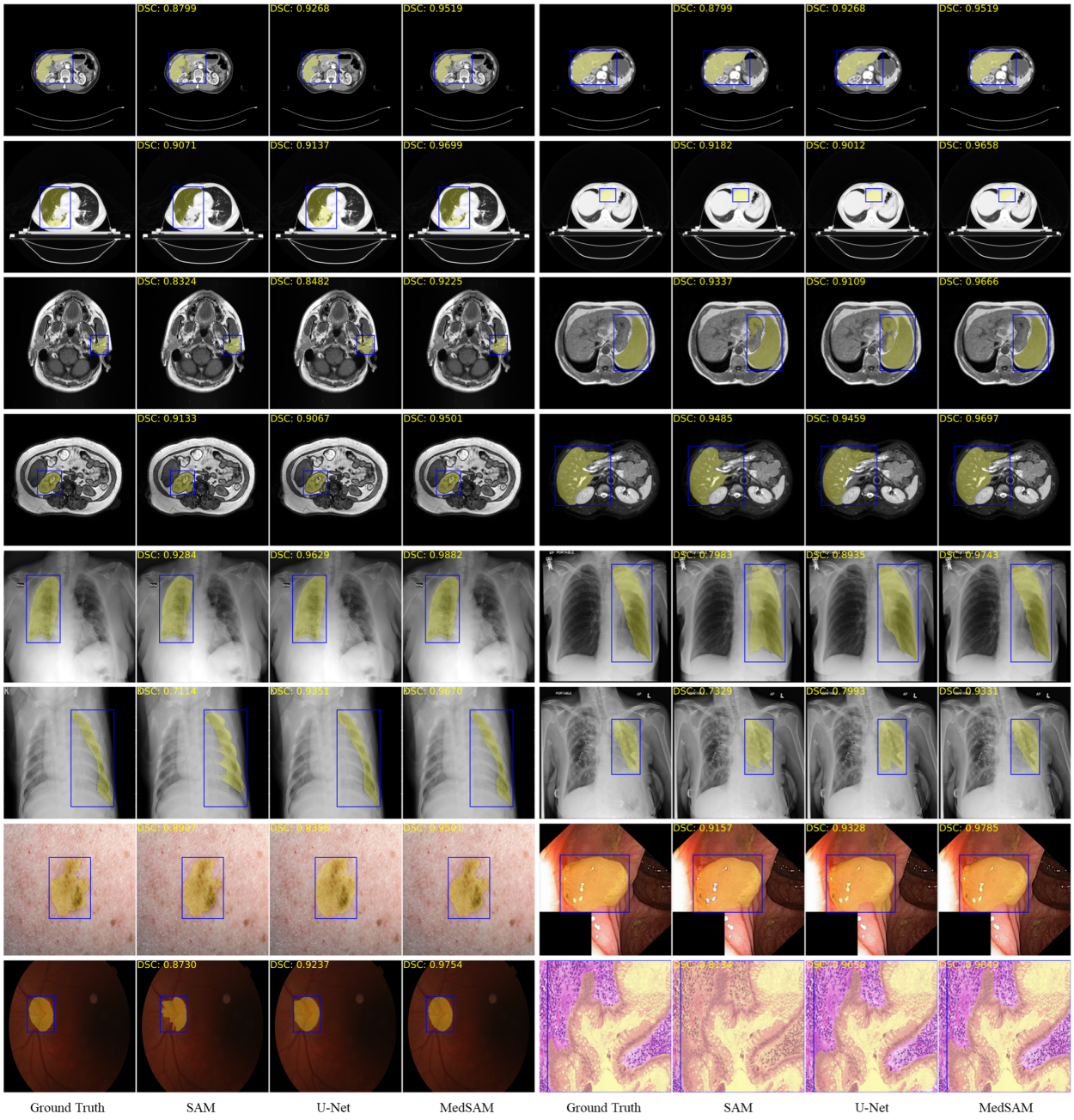

Fig. 10. Segmentation examples in external validation.

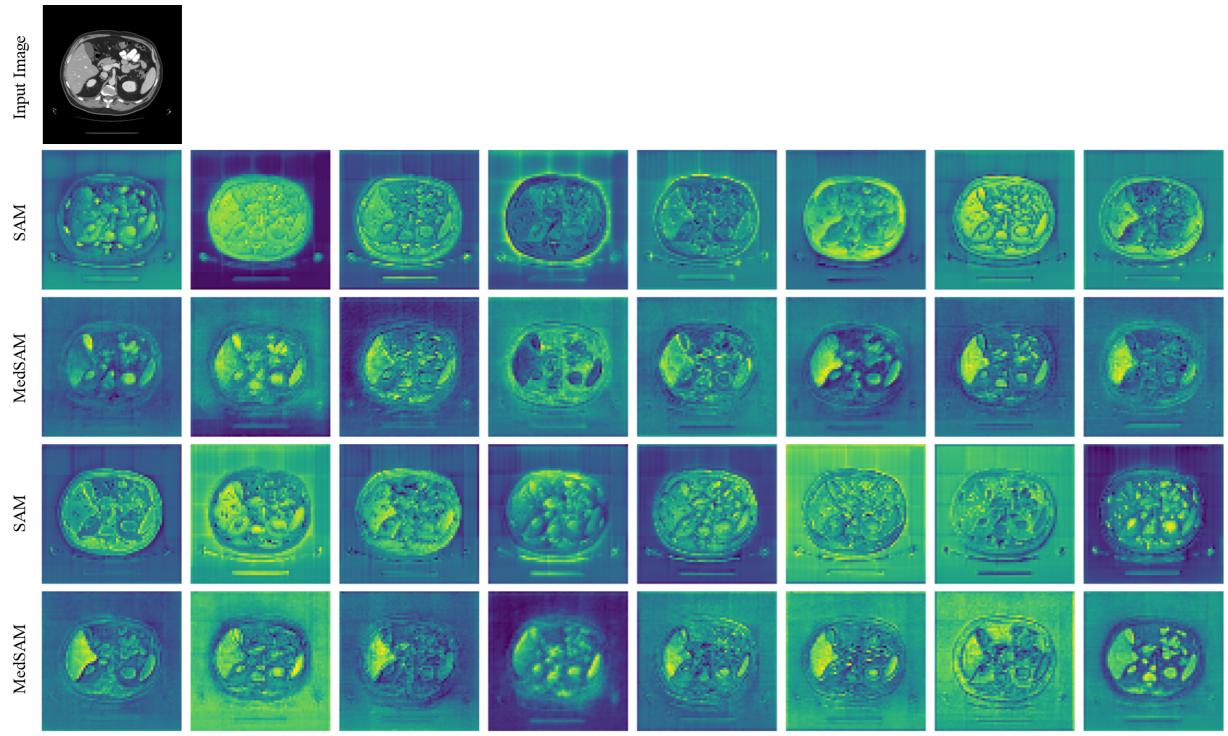

Fig. 11. Visualization and comparison of the saliency maps for image embeddings between SAM and MedSAM. MedSAM's features exhibited a greater abundance of semantic information, specifically pertaining to highly relevant anatomical structures.

TABLE 12
Data availability (CT and MR images).

Dataset	Modality	Download Link
AbdomenCT-1K	CT	https://github.com/JunMa11/AbdomenCT-1K
AMOS-CT	PET-CT	https://amos22.grand-challenge.org/
AutoPET	CT	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=93258287
COVID-19 Seg. Challenge	CT	https://covid-segmentation.grand-challenge.org/Data/
COVID-19-CT-Seg	CT	https://github.com/JunMa11/COVID-19-CT-Seg-Benchmark
GLIS-RT	CT	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=95224486
HCC-TACE-Seg	CT	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=70230229
HECKTOR	PET-CT	https://hecktor.grand-challenge.org/Overview/
INSTANCE	CT	https://instance.grand-challenge.org/
KiPA	CT	https://kipa22.grand-challenge.org/
KiTS	CT	https://kits-challenge.org/kits23/
Lymph Nodes	CT	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=19726546
MMWHS-CT	CT	https://zmiclab.github.io/zxh/0/mmwhs/
MSD-Colon Tumor	CT	http://medicaldecathlon.com/
MSD-Hepatic Tumor	CT	http://medicaldecathlon.com/
MSD-Lung Tumor	CT	http://medicaldecathlon.com/
MSD-Pancreas	CT	http://medicaldecathlon.com/
MSD-Spleen	CT	http://medicaldecathlon.com/
NSCLC Pleural Effusion	CT	https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=68551327
NSCLC Radiogenomics	CT	https://wiki.cancerimagingarchive.net/display/Public/NSCLC+Radiogenomics
ORG	CT	https://www.nature.com/articles/s41597-020-00715-8
SegTHOR	CT	https://competitions.codalab.org/competitions/21145
StructSeg	CT	https://structseg2019.grand-challenge.org/Dataset/
TotalSegmentator	CT	https://zenodo.org/record/6802614
WORD	CT	https://github.com/HiLab-git/WORD
ACDC	MR	https://humanheart-project.creatis.insa-lyon.fr/database/
AMOS-MR	MR	https://amos22.grand-challenge.org/Dataset/
ATLAS R2.0	MR	https://atlas.grand-challenge.org/
Brain Tumor Dataset Figshare	MR	https://www.kaggle.com/datasets/ashkhagan/figshare-brain-tumor-dataset
Brain TR-GammakKnife	MR	https://doi.org/10.7937/xb6d-py67
BraTS	MR	http://braintumorsegmentation.org/
CC-Tumor Heterogeneity	MR	https://doi.org/10.7937/ERZ5-QZ59
CHAOS	MR	https://chaos.grand-challenge.org/
crossMoDA	MR	https://crossmoda-challenge.ml/
FeTA	MR	https://feta.grand-challenge.org/
HaN-Seg	MR	https://zenodo.org/record/
ISLES	MR	http://www.isles-challenge.org/
I2CVB	MR	https://i2cvb.github.io/
Meningioma-SEG-CLASS	MR	https://doi.org/10.7937/0TKV-1A36
MMs	MR	https://www.ub.edu/mnms-2/
MMWHS-MR	MR	https://zmiclab.github.io/zxh/0/mmwhs/
MSD-Heart	MR	http://medicaldecathlon.com/
MSD-Prostate	MR	http://medicaldecathlon.com/#tasks
NCI-ISBI	MR	http://dx.doi.org/10.7937/K9/TCIA.2015.zF0vlOPv
PI-CAI	MR	http://github.com/DIAGNijmegen/picai_labels
PPMI	MR	https://www.ppmi-info.org/access-data-specimens/download-data
PROMISE	MR	https://promise12.grand-challenge.org/Details/
Qin-Prostate-Repeatability	MR	http://doi.org/10.7937/K9/TCIA.2018.MR1CKGND
QUBIQ	MR	https://qubiq21.grand-challenge.org/
Spine	MR	https://www.cg.informatik.uni-siegen.de/en/spine-segmentation-and-analysis
WMH	MR	https://wmh.isi.uu.nl/

TABLE 13
Data availability (Chest X-ray (CXR), Dermoscopy, Endoscopy, Fundus, Mammography, Optical Coherence Tomography (OCT), Pathology, Ultrasound images).

Dataset	Modality	Download Link
Chest Xray Masks and Labels	CXR	https://www.kaggle.com/datasets/nikhilpandey360/chest-xray-masks-and-labels
Chest X-Ray (Pneumothorax)	CXR	https://www.kaggle.com/datasets/vbookshelf/pneumothorax-chest-xray-images-and-masks
COVID-19 Radiography	CXR	https://www.kaggle.com/datasets/tawsifurrahman/covid19-radiography-database
COVID-QU-Ex	CXR	https://www.kaggle.com/datasets/anasmohammedtahir/covidqu
JSRT	CXR	http://imgcom.jsrt.or.jp/minijsrtdb/
Lung	CXR	https://doi.org/10.2214/ajr.174.1.1740071
QaTa-COV19	CXR	https://www.kaggle.com/datasets/aysendegerli/qatacov19-dataset
ISIC	Dermoscopy	https://challenge.isic-archive.com/data/
UWaterloo Skin Cancer	Dermoscopy	https://uwaterloo.ca/vision-image-processing-lab/research-demos/skin-cancer-detection
BKAI-IGH NeoPolyp	Endoscopy	https://www.kaggle.com/competitions/bkai-igh-neopolyp/data
CholecSeg8k	Endoscopy	https://www.kaggle.com/datasets/newslab/cholecseg8k
Kvasir	Endoscopy	https://datasets.simula.no/kvasir/
m2caiSeg	Endoscopy	https://www.kaggle.com/datasets/salmanmaq/m2caiseg
PolypGen	Endoscopy	https://www.synapse.org/#!Synapse:syn45200214
RobTool	Endoscopy	https://www.synapse.org/#!Synapse:syn22427422
sisvse	Endoscopy	https://www.kaggle.com/datasets/yjh4374/sisvse-dataset
IDRiD	Fundus	https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid
PAPILA	Fundus	https://figshare.com/articles/dataset/PAPILA/14798004/1
REFUGE	Fundus	https://refuge.grand-challenge.org/
CDD-CESM	Mammography	https://doi.org/10.7937/29kw-ae92
Intraretinal Cystoid Fluid	OCT	https://www.kaggle.com/datasets/zeeshanahmed13/intraretinal-cystoid-fluid
OCT Images (DME)	OCT	https://www.kaggle.com/datasets/paultimothymooney/chiu-2015
GlaS@MICCAI2015	Pathology	https://warwick.ac.uk/fac/cross_fac/tia/data/glascontest/download/
HuBMAP HPA	Pathology	https://www.kaggle.com/competitions/hubmap-organ-segmentation/
HuBMAP Hacking the Kidney	Pathology	https://www.kaggle.com/competitions/hubmap-kidney-segmentation/data?select=test
AbdomenUS	Ultrasound	https://www.kaggle.com/datasets/ignaciorlando/ussimandsegm
Breast Cancer	Ultrasound	https://www.kaggle.com/datasets/aryashah2k/breast-ultrasound-images-dataset
CAMUS	Ultrasound	https://www.creatis.insa-lyon.fr/Challenge/camus/
CT2USforKidneySeg	Ultrasound	https://www.kaggle.com/datasets/siatsyx/ct2usforkidneyseg
FH-PS-AOP	Ultrasound	https://ps-fh-aop-2023.grand-challenge.org/
HC	Ultrasound	https://hc18.grand-challenge.org/
TN-SCUI	Ultrasound	https://tn-scui2020.grand-challenge.org/
Nerve	Ultrasound	https://www.kaggle.com/competitions/ultrasound-nerve-segmentation/data

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